Articulate3D Challenge — Technical Report

Frozen USDNet, Geometric Priors, and Structural Post-Processing

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Team

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Final standing (boards frozen 2026-08-28): **1st** on Movable-dev ($AP_{50}^{axis+origin} = 0.5038$) and Interaction-dev ($AP_{50} = 0.6446$); **2nd** on Movable-test (0.4056) and Interaction-test (0.3286). A full write-up with all ablations and negative results is available on request ("Geometry Beats Regression: A Diagnosis-Driven, Almost-Training-Free Recipe for Scene-Level Articulated Part Understanding").

Method

Our recipe (Fig. 1) keeps the official USDNet baseline almost entirely frozen and improves the ranking metric $AP_{50}^{axis+origin} = AP_{50} \times axis \times origin$ by fixing the two articulation factors, which per-instance error forensics showed to fail for different reasons: the regressed *axis* collapses onto the dominant gravity-aligned mode, and the regressed *origin* shrinks toward the part centroid.

1. **Frozen segmentation.** Minkowski U-Net backbone, Mask3D decoder, mask head and class head are kept at the released train+val weights, *including BatchNorm running statistics*. AP_{50} is therefore bit-identical to the baseline within a checkpoint and every change is cleanly attributable.
2. **One trained component.** An 88 K-parameter attention-pooled origin head is finetuned with everything else frozen. Two silent pitfalls we hit and fixed: (a) frozen parameters do not stop BatchNorm buffers from drifting, which fakes gains on validation; (b) resuming a released checkpoint restarts USDNet's coarse-to-fine mask curriculum, which dilates the GT masks by 0.1 m for 100 epochs and destroys the model unless disabled.
3. **Training-free geometry overwrites the regressors.** Translation axis = PCA panel normal of the predicted mask, or in-plane if a sliding partner leaf exists; rotation axis forced vertical; rotation origin = mask endpoint perpendicular to the axis, on the side opposite the predicted handle (handles come from the companion interactable-part model). This alone takes test $AP_{50}^{axis+origin}$ from 0.264 to 0.3385 with the segmentation branch untouched.
4. **Structural post-processing (both tasks).** The model routinely merges the two leaves of a double door into one instance. Offline DBSCAN connected-component splitting + fragment filter + NMS, tuned with the official evaluator on validation, transfers nearly in full to test (movable $0.3385 \rightarrow 0.3760$; interaction $0.2686 \rightarrow 0.2886$).
5. **Deadline-day retrain and ensemble.** Because the released checkpoints were trained on train+val, validation is in-sample. A mirror-flip finetune (train+val, flip augmentation, curriculum off, LR 10^{-5}) gained only +0.004 on validation but +0.027 on test; an equal-weight ensemble of three such models (interaction) and the same recipe transplanted to the movable model gave the final 0.4056 / 0.3286.

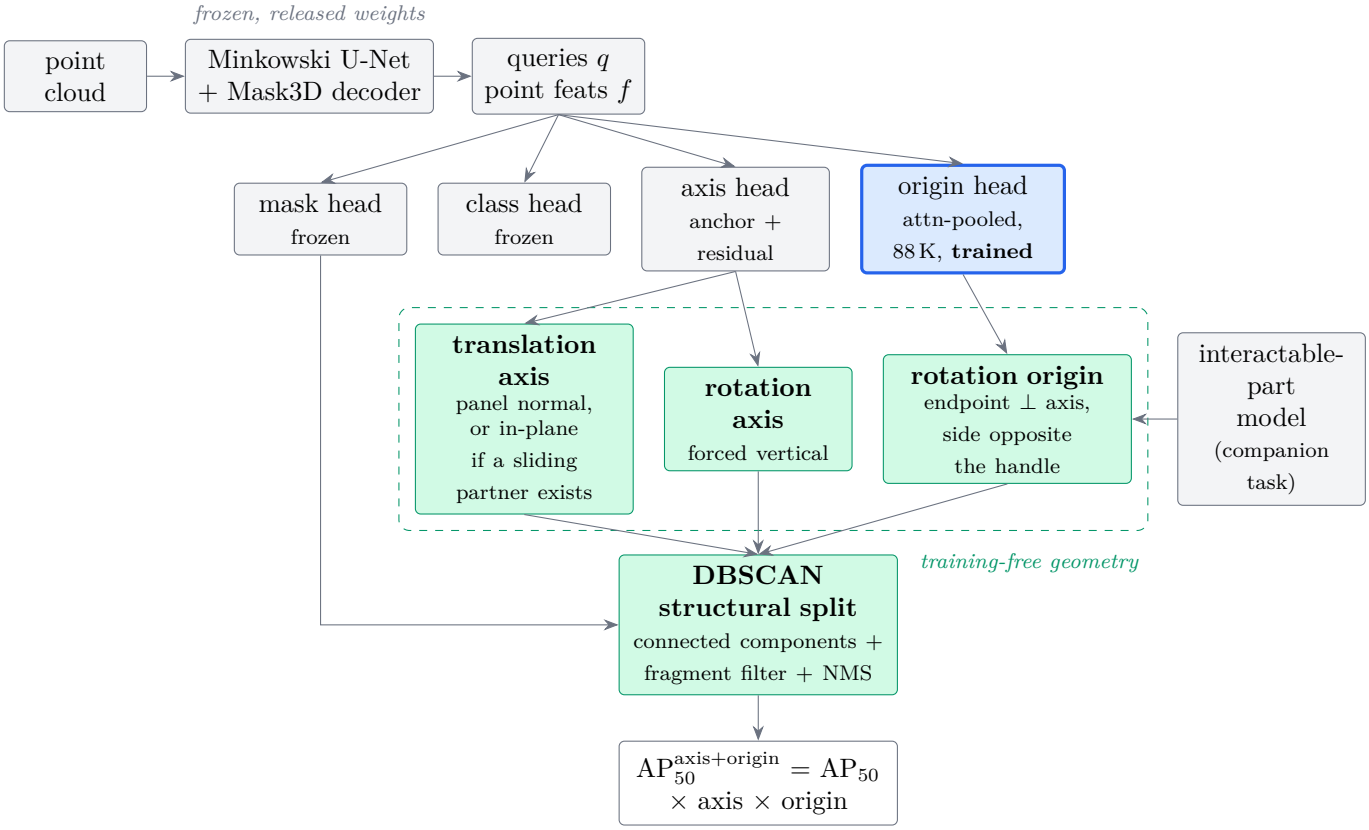


Figure 1: **Method overview.** Grey: frozen at released weights. Blue: the only trained component. Green: training-free geometry read off the predicted masks and the offline structural post-processing.

Results (challenge server, ranking metric)

Configuration (cumulative)	Movable test $AP_{50}^{axis+origin}$	Interaction test AP_{50}
USDNet baseline (reproduced)	0.2640	0.2686
+ trained origin head + geometric priors	0.3385	—
+ DBSCAN structural post-processing	0.3760	0.2886
+ flip finetune / ensemble (final)	**0.4056**	**0.3286**
Relative to baseline	+54%	+22%